Maths QB

Q.7)

→

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| X | 28 | 45 | 40 | 38 | 35 | 33 | 40 | 32 | 36 | 33 |
| Y | 23 | 34 | 33 | 34 | 30 | 26 | 28 | 31 | 36 | 35 |

→ Formulae

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \Sigma y^2}}$$

$$\bar{X} = \frac{360}{10} = 36$$

$$x = X - \bar{X}$$
$$y = Y - \bar{Y}$$

$$\bar{Y} = \frac{310}{10} = 31$$

| X | x<br>X - 36 | $x^2$ | Y | y<br>Y - 31 | $y^2$ | xy |
|---|---|---|---|---|---|---|
| 28 | -8 | 64 | 23 | -8 | 64 | 64 |
| 45 | 9 | 81 | 34 | 3 | 9 | 27 |
| 40 | 4 | 16 | 33 | 2 | 4 | 8 |
| 38 | 2 | 4 | 34 | 3 | 9 | 6 |
| 35 | -1 | 1 | 30 | -1 | 1 | 1 |
| 33 | -3 | 9 | 26 | -5 | 25 | 15 |
| 40 | 4 | 16 | 28 | -3 | 9 | -12 |
| 32 | -4 | 16 | 31 | 0 | 0 | 0 |
| 36 | 0 | 0 | 36 | 5 | 25 | 0 |
| 33 | -3 | 9 | 35 | 4 | 16 | -12 |
| 360 | | 216 | | | 162 | 97 |

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \Sigma y^2}} = \frac{97}{\sqrt{216 \cdot 162}} = \boxed{0.5186}$$

Q8)

| Price | 100 | 98 | 85 | 92 | 90 | 84 | 88 | 90 | 93 | 95 |
|---|---|---|---|---|---|---|---|---|---|---|
| sales | 500 | 610 | 700 | 630 | 670 | 800 | 800 | 750 | 700 | 690 |

→ Consider Price as x & sales as y

| x | $x$ $x-91.5$ | $x^2$ | y | y | $y^2$ | xy |
|---|---|---|---|---|---|---|
| 100 | 8.5 | 72.25 | 500 | -185 | 34225 | -1572.5 |
| 98 | 6.5 | 42.25 | 610 | 75 | 5625 | -487.5 |
| 85 | -6.5 | 42.25 | 700 | 15 | 225 | -97.5 |
| 92 | 0.5 | 0.25 | 630 | -55 | 3025 | -27.5 |
| 90 | -1.5 | 2.25 | 670 | -15 | 225 | 2.25 |
| 84 | -7.5 | 56.25 | 800 | 115 | 13225 | -862.5 |
| 88 | -3.5 | 12.25 | 800 | 115 | 13225 | -402.5 |
| 90 | -1.5 | 2.25 | 750 | 65 | 4225 | -97.5 |
| 93 | 1.5 | 2.25 | 700 | 15 | 225 | 22.5 |
| 95 | 3.5 | 12.25 | 69 | 5 | 25 | 17.5 |
| 915 | | 244.5 | 6850 | | 74250 | -3505.25 |

$\bar{x} = \frac{915}{10} = 91.5$ $\bar{y} = \frac{6850}{10} = 685$

$$r = \frac{\Sigma xy}{\sqrt{\Sigma x^2 \Sigma y^2}} = \frac{-3505.25}{\sqrt{244.5 \times 74250}}$$

$r = -0.8226$ //

Rank //

Q.9)

| | | | | | | |
|---|---|---|---|---|---|---|
| Marks in stat | 40 | 42 | 45 | 35 | 36 | 39 |
| Marks in Acc | 46 | 43 | 44 | 39 | 40 | 43 |

→ Consider Marks in stat as X &
Marks in Acc as Y

| X | Rank 1 $R_1$ | Y | Rank 2 $R_2$ | $D^2$ $(R_1 - R_2)^2$ |
|---|---|---|---|---|
| 40 | 4 | 46 | 6 | 4 |
| 42 | 5 | 43 | 3.5 | 2.25 |
| 45 | 6 | 44 | 5 | 1 |
| 35 | 1 | 39 | 1 | 0 |
| 36 | 2 | 40 | 2 | 0 |
| 39 | 3 | 43 | 3.5 | 0.25 |
| | | | | 7.5 |

$$r = 1 - \frac{6\left[\Sigma D^2 + \frac{1}{12}(m_1^3 - m_1)\right]}{n^3 - n}$$

$$r = 1 - \frac{6\left[7.5 + 0.5\right]}{(6)^3 - 6}$$

$$r = 1 - \frac{6\left[7.5 + 0.5\right]}{210}$$

$r = 0.77143$ //

10)

| X | 32 | 55 | 49 | 60 | 43 | 37 | 43 | 49 | 10 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|
| Y | 40 | 30 | 70 | 20 | 30 | 50 | 72 | 60 | 45 | 45 |

| → X | R1 | Y | R2 | $D^2$ $(R_1 - R_2)^2$ |
|---|---|---|---|---|
| 32 | 3 | 40 | 4 | 1 |
| 55 | 9 | 30 | 2.5 | 42.25 |
| 49 | 7.5 | 70 | 9 | 2.25 |
| 60 | 10 | 20 | 1 | 81 |
| 43 | 5.5 | 30 | 2.5 | 9 |
| 37 | 4 | 50 | 7 | 9 |
| 43 | 5.5 | 72 | 10 | 20.25 |
| 49 | 7.5 | 60 | 8 | ~~0.5~~ 0.25 |
| 10 | 1 | 45 | 5.5 | 20.25 |
| 20 | 2 | 45 | 5.5 | 12.25 |
| | | | | 197.5 |

$$r = 1 - 6\left[\frac{\Sigma D^2 + \frac{1}{12}(m_1^3 - m_1) + \frac{1}{12}(m_2^3 - m_2) + \frac{1}{2}(m_3^3 - m_3)}{n^3 - n}\right]$$

$$= 1 - 6\left[\frac{\cancel{197.5}\ 197.5 + 0.5 + 0.5 + 0.5}{(10)^3 - 10}\right]$$

$$= 1 - 6\left[\frac{199}{990}\right]$$

$$r = -0.20606$$ //

Q.11) Equations of two regression line, are $3x + 2y = 26$ & $6x + y = 31$ then Find $r$.

→ ∴ given equations are

$3x + 2y = 26$ & (1)

& $6x + y = 31$ (2)

∴ Express both equation in standard regression form

∴ $3x + 2y = 26$

$2y = 26 - 3x$

$y = \frac{26}{2} - \frac{3x}{2}$

$y = 13 - 1.5x$

∴ Slope $\boxed{b_{yx} = -1.5}$

∴ $6x + y = 31$

~~$y = 31 - 6x$~~ $6x = 31 - y$

$x = \frac{31}{6} - \frac{1}{6}y$

∴ Slope $\boxed{b_{xy} = -\frac{1}{6}}$

∴ By using regression slope relationship

$r^2 = b_{yx} \times b_{xy}$

$$r^2 = (-1.5) \times \left(-\frac{1}{6}\right) = \frac{1.5}{6} = \frac{1}{4}$$

$$\boxed{r = \pm \frac{1}{2}}$$

Determine the sign of $r$

we know both slopes are negative, which means as $x$ increases, $y$ decreases so, $r$ is negative.

$$\boxed{r = -\frac{1}{2}} \quad \therefore \boxed{r = -0.5}$$

12] regression lines are $2x - y = 5$ & $y - 3x - 8 = 0$.

→

given equations are.

$$2x - y = 5 \quad (1)$$
$$y - 3x - 8 = 0$$
$$y - 3x = 8 \quad (2)$$

Expressing both equation is standard regression form

$$2x - y = 5$$
$$-y = 5 - 2x$$
$$y = -5 + 2x$$

$\therefore$ slope $\boxed{b_{yx} = 2}$

$$y - 3x = 8$$
$$-3x = 8 - y$$
$$3x = -8 + y$$
$$x = \frac{-8}{3} + \frac{1}{3}y$$

$$\therefore \boxed{b_{xy} = \frac{1}{3}}$$

$\therefore$ by using regression slope relationship

$$r^2 = b_{yx} \times b_{xy}$$
$$r^2 = 2 \times \frac{1}{3}$$
$$r = \sqrt{2 \times \frac{1}{3}}$$
$$r = \pm 0.816$$

Now to solve means $\bar{x}$ & $\bar{y}$

$\therefore$ we know

$$\cancel{2x - y = 5} \quad \cancel{(1)}$$
$$\& \quad \cancel{-3x + y = 8} \quad \cancel{(2)}$$

$$2x - y = 5$$
$$\therefore \quad y = 2x - 5 \quad (3)$$
$$\& \quad y - 3x = 8$$
$$y = 3x + 8 \quad (4)$$

$$\therefore \quad y = 2x - 5$$
$$y = 3x + 8$$
$$\therefore \quad 2x - 5 = 3x + 8$$

$\therefore -5 - 8 = 3x - 2x$

$\therefore -13 = x$

$\boxed{\bar{x} = -13}$

substutating $\bar{x}$ in (3) we get.

$\therefore y = 3x + 8$

$y = 3(-13) + 8$

$\boxed{y = -31}$

$\therefore$ means $\boxed{\bar{x} = -13 \quad \& \quad \bar{y} = -31}$

regression

13]

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| X | 36 | 32 | 34 | 31 | 32 | 32 | 34 |
| Y | 35 | 33 | 31 | 30 | 34 | 32 | 36. |

→

| X | x<br>X − 23.1 | $x^2$ | Y | y<br>Y − 23.1 | $y^2$ | xy |
|---|---|---|---|---|---|---|
| 36 | 12.9 | 166.41 | 35 | 11.9 | 141.61 | 153.51 |
| 32 | 8.9 | 79.21 | 33 | 9.9 | 98.01 | 88.11 |
| 34 | 10.9 | 118.81 | 31 | 7.9 | 62.41 | 86.11 |
| 31 | 7.9 | 62.41 | 30 | 6.9 | 47.61 | 54.51 |
| 32 | 8.9 | 79.21 | 34 | 10.9 | 118.81 | 97.01 |
| 32 | 8.9 | 79.21 | 32 | 8.9 | 79.21 | 79.21 |
| 34 | 10.9 | 118.81 | 36 | 12.9 | 166.41 | 140.61 |
| | | 704.07 | | | 714.07 | 699.07. |

$\bar{X} = \frac{231}{10} = 23.1$ $\bar{Y} = \frac{231}{10} = 23.1$

$b_{yx} = \frac{\Sigma xy}{\Sigma x^2}$

$b_{yx} = \frac{699.07}{704.07}$

$b_{yx} = 0.9928$

$b_{xy} = \frac{\Sigma xy}{\Sigma y^2}$

$b_{xy} = \frac{699.07}{714.07}$

$b_{xy} = 0.9789$ //

$y - \bar{Y} = b_{yx} \cdot (x - \bar{x})$

$Y - 23.1 = 0.9928(x - 23.1)$

$Y = 0.9928x - 22.93368 + 23.1$

$Y = 0.9928x + 0.16632$ //

$\cancel{(x-x)}$ $x - \bar{x} = b_{xy}(y - \bar{y})$

$x - 23.1 = 0.9789(y - 23.1)$

$x - 23.1 = 0.9789y - 22.61259$

$x = 0.9789y - 0.48741$ //

$\therefore X = 0.9789y + 0.4874$ &

$Y = 0.9928x + 0.16632$ //

14)

| x | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| y | 1 | 5 | 11 | 8 | 14 |

straight line

→ consider the equation of line

$y = a + bx$

then normal equations are.

$\Sigma y = na + b\Sigma x$

$\Sigma xy = a\Sigma x + b\Sigma x^2$

| $x$ | $x^2$ | $y$ | | $xy$ |
|---|---|---|---|---|
| 1 | 1 | 1 | | 1 |
| 2 | 4 | 5 | | 10 |
| 3 | 9 | 11 | | 33 |
| 4 | 16 | 8 | | 32 |
| 5 | 25 | 14 | | 70 |
| 15 | 55 | 39 | | 146 |

$\therefore \Sigma y = na + b\Sigma x$

$\cancel{39 = 5a + 15b}$ ... (1)

$39 = 5a + 15b$ ... ①

$\therefore \Sigma xy = a\Sigma x + b\Sigma x^2$

$146 = 15a + 55b$ ②

$\therefore$ Solving equation ① & ② we get.

$\therefore a = -0.9$

$b = 2.9$

$\therefore$ Substutating $a = -0.9$ & $b = 2.9$ in equation of line.

$\therefore y = a + bx$

$\therefore y = -0.9 + 2.9x$

$\therefore \boxed{y = 2.9x - 0.9}$ //

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15] 

| Year (x) | 1951 | 1961 | 1971 | 1981 | 1991 |
|---|---|---|---|---|---|
| Productivity (y) | 10 | 12 | 8 | 10 | 15 |

consider the equation of line

$y = a + bx$

$\therefore$ the normal equations are.

$\Sigma y = na + b\Sigma x$

$\Sigma xy = a\Sigma x + b\Sigma x^2$

| $x$ | $x^2$ | $y$ | $xy$ |
|---|---|---|---|
| 1951 | 3806401 | 10 | 19510 |
| 1961 | 3845521 | 12 | 23532 |
| 1971 | 3884841 | 8 | 15768 |
| 1981 | 3924361 | 10 | 19810 |
| 1991 | 3964081 | 15 | 29865 |
| 9855 | 19425205 | 55. | 108485. |

$\therefore \Sigma y = na + b\Sigma x$

$\therefore 55 = 5a + 9855b \quad \ldots (1)$

$\therefore \Sigma xy = \Sigma x a + b\Sigma x^2$

$\therefore 108485 = 9855a + 19425205b. \quad (2)$

Solving equation (1) & (2) we get.

$\therefore a = -146.68$

$b = 0.08.$

$\therefore$ Substutating a & b in equation of line.

$\therefore y = a + bx$

$y = -146.68 + 0.08x$

$y = 0.08x - 146.68.$

$\therefore$ Now we need to find production in 1987.

$\therefore x = 1987$.

$\therefore$ substutating $x$ in equation of line we get.

$$y = 0.08x - 146.68$$
$$y = 0.08(1987) - 146.68$$
$$y = 158.96 - 146.68$$
$$\boxed{y = 12.28.}$$

$\therefore$ the estimate production ~~of~~ in 1987 is 12.28.

$0.14652$

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(Probability)

~~16)~~

16) Find probability that at must 4 ~~different~~ defective bulbs will be found in box of 200 bulbs if it is known that 2 percent of bulbs are defective

→

∴ to calculate mean

$m = n * p \qquad n = 200$

$m = 200 \times 0.02 \qquad p = 2\% = 0.02$

∴ $m = 4$

∴ Using poison probability formulae for $x = 0, 1, 2, 3, 4$

∴ calculating $P(x=0) = \dfrac{e^{-4}\, 4^0}{0!} = \boxed{0.01831}$

∴ $P(x=1) = \dfrac{e^{-4}\, 4^1}{1!} = 4 \times 0.01831 = 0.07326$

∴ For $P(x \le 4) = P(x=0) + P(x=1) + P(x=2) + P(x=3) + P(x=4)$

$= 0.01831 + 0.07326 + \dfrac{e^{-4}\, 4^2}{2!} + \dfrac{e^{-4}\, 4^3}{3!} + \dfrac{e^{-4}\, 4^4}{4!}$

$= 0.01831 + 0.07326 + 0.14652 + 0.19536 + 0.19536$

$\boxed{P(x \le 4) = 0.62881}$ //

Poision distribution

17) → given.

$p = 0.01$

$n = 10$

$m = n \cdot p$

$\therefore m = 10(0.01$

$\boxed{m = 0.1}$ //

$\therefore P(x = r) = \frac{e^{-m} m^{x}}{x!}$

$P(x > 1) = 1 - P(X < 1)$

$= 1 - P(x = 0) - P(x = 1)$.

$P(x > 1) = 1 - \frac{e^{-0.1}(0.1)^0}{0!} - \frac{e^{-0.1}(0.1)^1}{1!}$

$P(x > 1) = 1 - 0.9048 - 0.0905$

$P(x > 1) = 0.0047$ //

The probability of more than 1 error in 10 recieved digits is 0.0047 //